

PAPER_750: M51 Whirlpool and NGC 1316 Fornax A — MUGE with Simulation Scripts

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Abstract

Two Hubble-studied galaxies — M51 (the Whirlpool Galaxy, interacting with NGC 5195) and NGC 1316 (Fornax A, a post-merger elliptical) — represent opposite ends of the galaxy evolution spectrum. This paper derives tailored Master Universal Gravity Equations (MUGEs) for both systems using Hubble ACS datasets, incorporating interaction-specific environmental terms: F_{tidal} and ψ_{spiral} for M51, and $F_{\text{tidal}} + F_{\text{cluster}} + \rho_{\text{dust}}$ for NGC 1316. Full Python simulation scripts (m51_simulation.py, ngc1316_simulation.py) are included for radial acceleration profile generation.

1. Introduction

1.1 M51 (Whirlpool Galaxy)

M51 is a grand design spiral at 7.7 Mpc with an active tidal interaction companion NGC 5195. Hubble ACS observations (2005) provide: - $M_{\text{visible}} = 1.2 \times 10^{11} M_{\odot}$ - $M_{\text{DM}} = 4 \times 10^{10} M_{\odot}$

- $\text{SFR} \approx 1 M_{\odot}/\text{yr}$ - $M_{\text{BH}} \approx 10^6 M_{\odot}$ - Distance = 7.7 Mpc, $z \approx 0.0015$ - $B \approx 5 \times 10^{-6} \text{ T}$

1.2 NGC 1316 (Fornax A)

NGC 1316 is a giant elliptical at 75 Mpc with a complex merger history: - $M_{\text{total}} \approx 5 \times 10^{11} M_{\odot}$ (visible + DM) - $M_{\text{DM}} = 1.5 \times 10^{11} M_{\odot}$ - $M_{\text{BH}} \approx 10^8 M_{\odot}$ - Merger age: 1-3 Gyr ago (from ripples, dust lanes) - $\rho_{\text{dust}} \approx 10^{-21} \text{ kg/m}^3$

2. M51 Whirlpool Galaxy MUGE

$$\begin{aligned} g_{\text{M51}}(r, t) = & (G \cdot M(t)) / (r(t)^2) \cdot (1 + H(t, z)) \cdot (1 - B(t)/B_{\text{crit}}) \cdot (1 + F_{\text{env}}(t)) \\ & + (U_{\text{g1}} + U_{\text{g2}} + U_{\text{g3}'} + U_{\text{g4}}) + U_{\text{i}} \\ & + (\Lambda \cdot c^2 / 3) \\ & + (\hbar / \sqrt{(\Delta x \cdot \Delta p)}) \cdot \int (\psi_{\text{total}} \cdot \hbar \cdot \psi_{\text{total}} dV) \cdot (2\pi / t_{\text{Hubble}}) \\ & + \rho_{\text{fluid}} \cdot V \cdot g \\ & + (M_{\text{vis}} + M_{\text{DM}}) \cdot (\delta \rho / \rho + (3 \cdot G \cdot M) / r^3) \end{aligned}$$

M51 F_{env} Terms

$$F_{\text{env}}(t)_{\text{M51}} = F_{\text{tidal}} + \psi_{\text{spiral_correction}}$$

$$F_{\text{tidal}} = G \cdot M_{\text{NGC5195}} / d_{\text{interaction}}^2$$

$$M_{\text{NGC5195}} \approx 10^{10} M_{\odot} = 1.989 \times 10^{40} \text{ kg}$$

$$d_{\text{interaction}} \approx 50 \text{ kpc} = 1.54 \times 10^{21} \text{ m}$$

$$F_{\text{tidal}} = 6.674 \times 10^{-11} \times 1.989 \times 10^{40} / (1.54 \times 10^{21})^2$$

$$F_{\text{tidal}} \approx 5.59 \times 10^{-13} \text{ m/s}^2 \quad (\text{normalized to } F_{\text{env}} \text{ fraction})$$

$$\psi_{\text{spiral}} = A \cdot e^{(-r^2/(2\sigma^2))} \cdot e^{i(m \cdot \theta - \omega \cdot t)}$$

$\sigma \approx 1 \text{ kpc} = 3.086 \times 10^{19} \text{ m}$ (spiral arm width)

$m = 2$ (grand design two-arm spiral)

M51 H(t,z)

$$H(t,z) = H_0 \cdot \sqrt{(0.3 \cdot (1 + 0.0015)^3 + 0.7)} \approx H_0 \cdot \sqrt{0.7003} \approx H_0$$

$z = 0.0015$ (negligible correction)

M51 Parameters Summary

Parameter	Value
M_visible	$1.2 \times 10^{11} M_{\odot} = 2.39 \times 10^{41} \text{ kg}$
M_DM	$4 \times 10^{10} M_{\odot} = 7.96 \times 10^{40} \text{ kg}$
r_galaxy	$30 \text{ kpc} = 9.26 \times 10^{20} \text{ m}$
B	$5 \times 10^{-6} \text{ T}$
SFR	$1 M_{\odot}/\text{yr} = 6.3 \times 10^{22} \text{ kg/yr}$
F_tidal	$5.59 \times 10^{-13} \text{ m/s}^2$

3. M51 Simulation Script (m51_simulation.py)

```
import numpy as np
import matplotlib.pyplot as plt

# Constants
G = 6.6743e-11          # m^3 kg^-1 s^-2
H_0 = 70e3 / 3.086e22  # s^-1
Lambda = 1.1e-52        # m^-2
c = 3e8                  # m/s
hbar = 1.0546e-34       # J.s
t_Hubble = 4.35e17      # s
B_crit = 1e11            # T
rho_vac_ScM = 7.09e-37  # J/m^3
rho_vac_UA = 7.09e-36   # J/m^3
mu_0 = 4 * np.pi * 1e-7 # H/m
M_sun = 1.989e30         # kg
kpc = 3.086e19          # m

# M51 Parameters
M_vis = 1.2e11 * M_sun
M_DM = 4.0e10 * M_sun
M_NGC5195 = 1e10 * M_sun
r_0 = 30e3 * kpc
d_inter = 50e3 * kpc
B = 5e-6              # T
sigma_spiral = 1e3 * kpc
z = 0.0015
lambda_I = 1.0
```

```

F_RZ = 0.01
omega_i = 1.585e-8      # rad/s

def M_total():
    return M_vis + M_DM

def H_z(z):
    return H_0 * np.sqrt(0.3 * (1 + z)**3 + 0.7)

def F_tidal():
    return G * M_NGC5195 / d_inter**2

def U_i():
    return lambda_I * (rho_vac_SCm / rho_vac_UA) * omega_i * 1.0 * (1 + F_RZ)

def psi_spiral(r):
    A = 1e-10
    return A * np.exp(-r**2 / (2 * sigma_spiral**2))

def g_M51(r, t=0):
    M = M_total()
    H = H_z(z)
    F_env = F_tidal() / (G * M / r**2) # normalized
    g_grav = (G * M / r**2) * (1 + H) * (1 - B / B_crit) * (1 + F_env)
    quantum = hbar / np.sqrt(1e-10 * 1e-20) * psi_spiral(r)**2 * (2 * np.pi / t_Hubble)
    dm_term = (M_vis + M_DM) * (1e-5 + 3 * G * M / r**3)
    cosmological = Lambda * c**2 / 3
    return g_grav + U_i() + cosmological + quantum + dm_term

# Compute radial profile
r_vals = np.linspace(1e3 * kpc, 30e3 * kpc, 200)
g_vals = [g_M51(r) for r in r_vals]

plt.figure(figsize=(10, 6))
plt.plot(r_vals / kpc, g_vals, 'b-', linewidth=2, label='M51 MUGE Acceleration')
plt.xlabel('Radius (kpc)')
plt.ylabel('g_M51 (m/s²)')
plt.title('M51 Whirlpool Galaxy – UQFF MUGE Radial Profile')
plt.legend()
plt.grid(True, alpha=0.3)
plt.savefig('m51_gravity_profile.png', dpi=150)
plt.close()
print("M51 simulation complete. Profile saved to m51_gravity_profile.png")

```

4. NGC 1316 (Fornax A) MUGE

$$\begin{aligned}
g_{\text{NGC1316}}(r, t) = & (G \cdot M(t)) / (r(t)^2) \cdot (1 + H(t, z)) \cdot (1 - B(t)/B_{\text{crit}}) \cdot (1 + F_{\text{env}}(t)) \\
& + (U_{g1} + U_{g2} + U_{g3} + U_{g4}) + U_i \\
& + (\Lambda \cdot c^2 / 3) \\
& + (\hbar / \sqrt{(\Delta x \cdot \Delta p)}) \cdot \int (\psi_{\text{total}} \cdot H \cdot \psi_{\text{total}} dV) \cdot (2\pi / t_{\text{Hubble}}) \\
& + \rho_{\text{dust}} \cdot V \cdot g \\
& + (M_{\text{vis}} + M_{\text{DM}}) \cdot (\delta\rho/\rho + (3 \cdot G \cdot M)/r^3)
\end{aligned}$$

NGC 1316 F_env Terms

$$F_{\text{env}}(t)_{\text{NGC1316}} = F_{\text{tidal}} + F_{\text{cluster}}$$

F_{tidal} : tidal from past spiral galaxy mergers

$$F_{\text{tidal}} = G \cdot M_{\text{spiral}} / d_{\text{spiral}}^2$$

$$M_{\text{spiral}} \approx 10^{10} M_{\odot} \text{ (remnant progenitor)}$$

$$d_{\text{spiral}} \approx 50 \text{ kpc}$$

$$F_{\text{tidal}} \approx 5.59 \times 10^{-13} \text{ m/s}^2$$

F_{cluster} : star cluster tidal disruption

$$F_{\text{cluster}} = k_{\text{cluster}} \cdot M_{\text{cluster}}$$

$$k_{\text{cluster}} \approx 10^{-12} \text{ N/M}_{\odot}$$

$$M_{\text{cluster}} \approx 10^6 M_{\odot}$$

$$F_{\text{cluster}} \approx 10^{-6} \text{ m/s}^2$$

ρ_{dust} : dust lane fluid term

$$\rho_{\text{dust}} = 10^{-21} \text{ kg/m}^3 \text{ (ACS-derived dust column density)}$$

NGC 1316 Mass Evolution

$$M(t) = M_{\text{vis}} + M_{\text{DM}} + M_{\text{merger}}(t)$$

$$M_{\text{merger}}(t) = M_{\text{spiral}} \cdot e^{(-t/\tau_{\text{merge}})}$$

$$\tau_{\text{merge}} \approx 1 \text{ Gyr} = 3.156 \times 10^{16} \text{ s}$$

5. NGC 1316 Simulation Script (ngc1316_simulation.py)

```
import numpy as np
import matplotlib.pyplot as plt

# Constants (same as M51)
G = 6.6743e-11
H_0 = 70e3 / 3.086e22
Lambda = 1.1e-52
c = 3e8
hbar = 1.0546e-34
t_Hubble = 4.35e17
B_crit = 1e11
rho_vac_SCm = 7.09e-37
mu_0 = 4 * np.pi * 1e-7
M_sun = 1.989e30
kpc = 3.086e19

# NGC 1316 Parameters
M_vis = 3.5e11 * M_sun
M_DM = 1.5e11 * M_sun
M_spiral = 1e10 * M_sun
M_BH = 1e8 * M_sun
tau = 3.156e16          # 1 Gyr merger decay
d_spiral = 50e3 * kpc
rho_dust = 1e-21        # kg/m^3
B = 1e-4                # T
z = 0.005               # NGC 1316 redshift
```

```

k_cluster = 1e-12          # N/M_sun
M_cluster = 1e6 * M_sun
sigma_dust = 2e3 * kpc

def M_total(t):
    return M_vis + M_DM + M_spiral * np.exp(-t / tau)

def H_z(z):
    return H_0 * np.sqrt(0.3 * (1 + z)**3 + 0.7)

def F_env(t):
    F_tidal = G * M_spiral / d_spiral**2
    F_cluster = k_cluster * M_cluster
    return F_tidal + F_cluster

def U_i():
    rho_vac_UA = 7.09e-36
    return 1.0 * (rho_vac_SCm / rho_vac_UA) * 1e-8 * 1.0 * 1.01

def g_NGC1316(r, t=3.156e15): # t=100 Myr default
    M = M_total(t)
    H = H_z(z)
    F = F_env(t) / (G * M / r**2)
    g_grav = (G * M / r**2) * (1 + H) * (1 - B / B_crit) * (1 + F)
    g_dust = rho_dust * (4/3 * np.pi * r**3) * G * M / r**2
    dm_term = (M_vis + M_DM) * (1e-5 + 3 * G * M / r**3)
    cosmological = Lambda * c**2 / 3
    return g_grav + U_i() + cosmological + g_dust + dm_term

# Compute
r_vals = np.linspace(1e3 * kpc, 100e3 * kpc, 100)
g_vals = [g_NGC1316(r) for r in r_vals]

plt.figure(figsize=(10, 6))
plt.plot(r_vals / kpc, g_vals, 'r-', linewidth=2, label='NGC 1316 MUGE Acceleration')
plt.xlabel('Radius (kpc)')
plt.ylabel('g_NGC1316 (m/s²)')
plt.title('NGC 1316 Fornax A – UQFF MUGE Radial Profile')
plt.legend()
plt.grid(True, alpha=0.3)
plt.savefig('ngc1316_gravity_profile.png', dpi=150)
plt.close()
print("NGC 1316 simulation complete. Profile saved to ngc1316_gravity_profile.png")

```

6. Comparative Analysis

Property	M51 (Whirlpool)	NGC 1316 (Fornax A)
Type	Sc spiral	E/S0 elliptical
Distance	7.7 Mpc	75 Mpc
Interaction	Active (NGC 5195)	Past mergers (~1-3 Gyr)
Key term	F_tidal (live)	F_cluster + ρ_{dust}

Property	M51 (Whirlpool)	NGC 1316 (Fornax A)
Star formation	$\text{SFR} \approx 1 \text{ M}_{\odot}/\text{yr}$	~ 0 (quenched)
AGN activity	Moderate	Strong (radio lobes)
Key physics	Density waves (ψ_{spiral})	Dust lane drag (ρ_{dust})

7. Astrophysical Significance

M51: The F_{tidal} term drives the prominent spiral arms visible in Hubble observations. The ψ_{spiral} density wave modulation affects star formation efficiency across the disk. NGC 5195's tidal force creates the bridge structure and enhanced SFR in the outer arms.

NGC 1316: The F_{cluster} term explains the deficit of low-mass globular clusters in the inner region (disrupted by tidal forces during multiple mergers). The ρ_{dust} dust lane creates a complex hydrodynamic environment near the AGN.

8. Conclusion

The M51 and NGC 1316 MUGEs demonstrate the UQFF framework's ability to model both active interaction (M51) and post-merger (NGC 1316) galaxy evolution with Hubble-calibrated parameters. The included Python simulation scripts provide quantitative radial acceleration profiles ready for comparison with observational rotation curves and X-ray surface brightness profiles. These two contrasting systems validate the modularity of $F_{\text{env}}(t)$ across diverse galactic environments.

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